

IEEE 5G Core Testbed Administrator Guide

Package Generation, License Issuance & Deployment Management

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1. Overview

This guide is for system administrators responsible for issuing licenses and deploying the IEEE 5G Core Testbed platform. The platform is built on Open5GS running on k3s Kubernetes, managed via a web-based UI. Licenses are cluster-locked using JWT tokens signed with RSA-2048.

2. Architecture Overview

The license issuance architecture follows a hub-and-spoke model. The administrator holds the RSA-2048 private key and signs all license JWTs. Each deployment site receives a self-contained package and a site-specific license. The backend validates the license on every startup.

Component	Description
Admin (you)	Holds RSA-2048 private key — signs license JWTs
Deployment package	ieee_5g_core_deploy_v{version}.tar.gz (~5 KB)
License file	{site}-license.jwt — cluster-locked JWT
bootstrap.sh	Clones from GitHub, runs install.sh, places license
Backend startup check	Verifies JWT signature + k3s cluster UID on every boot

3. Prerequisites

Requirement	Details
Node.js	v18 or later (for generate-license.js)
OpenSSL	For key pair generation
Git	Access to ieee-testbed/ieee_5g_core repo
Private key	scripts/license-private.pem (never share)

4. One-Time Setup: Generate RSA Key Pair

Run once. The private key must be kept secure — never committed to git, never shared.

```
cd ieee_5g_core
bash scripts/generate-keypair.sh
```

Output — two files are created:

- **scripts/license-private.pem** → KEEP SECRET. Store in secure vault (e.g. 1Password, AWS Secrets Manager)
- **ui/backend/src/license-public.pem** → Commit to repository. Bundled in the backend binary.

■ **Warning: The private key cannot be recovered if lost. Losing it means you cannot issue new licenses. Back it up immediately in a secure location.**

5. Issuing a License

Step 1 — Receive cluster UID from site

Ask the site admin to run the following command on their server AFTER k3s is installed, and send you the output:

```
kubectl get namespace kube-system \
-o jsonpath='{.metadata.uid}'
```

Example output: a3f7c821-4b2e-11ef-9a1d-00163e002a4b

Step 2 — Generate the license

```
node scripts/generate-license.js \
--cluster-id a3f7c821-4b2e-11ef-9a1d-00163e002a4b \
--institution "University of Sfax" \
--features multus, tracing, gnb_mgmt \
--max-gnbs 10 \
--days 365 \
--out sfax-license.jwt
```

Feature flags:

Feature Flag	Enables
multus	Networking / Multus CNI management tab
tracing	Packet capture & NGAP tracing
gnb_mgmt	gNB registry, SCTP association view
all	All current and future features

Step 3 — Deliver to site

Send the following two files to the site administrator via secure channel (encrypted email or secure file share):

- ieee_5g_core_deploy_v{version}.tar.gz
- {site}-license.jwt

6. Building the Deployment Package

```
cd ieee_5g_core
bash scripts/package.sh
# Output: ieee_5g_core_deploy_v{git-tag}.tar.gz
```

The package contains only bootstrap.sh and README.txt (~5 KB). All Kubernetes manifests and application code are pulled from GitHub at deploy time. Ensure the repo is accessible from the deployment site.

7. License Renewal

Licenses expire based on the **--days** parameter. The backend warns 30 days before expiry in the UI footer. To renew:

1. Run generate-license.js with the same **cluster-id** and a new **--days** value
2. Send the new license.jwt to the site
3. Site admin replaces /etc/ieee5g/license.jwt and restarts the backend:

```
sudo cp new-license.jwt /etc/ieee5g/license.jwt
kubectl rollout restart deployment/core-ui -n open5gs
```

8. Revoking a License

To revoke access, simply do not renew the license when it expires. If immediate revocation is needed:

- Issue a replacement license with **--days 0** (expires immediately)
- Or: contact the site to delete **/etc/ieee5g/license.jwt** and restart core-ui

9. Tracking Deployments

Maintain a deployment registry (spreadsheet or CRM) with the following fields:

Field	Example
Institution	University of Tunis, Sfax, ...
Cluster UID	a3f7c821-4b2e-11ef-9a1d-00163e002a4b
Issued date	2026-03-18
Expiry date	2027-03-18
Features	multus, tracing
Max gNBs	10
License file	sfax-license.jwt

10. Troubleshooting

Problem	Likely Cause	Fix
"License not found"	File missing	Place license.jwt at /etc/ieee5g/license.jwt
"Invalid signature"	Wrong public key in build	Regenerate keypair, rebuild backend
"Not valid for this cluster"	Wrong cluster UID	Re-issue license with correct UID
"License expired"	Past expiry date	Issue renewal license
API returns 403	Backend in unlicensed mode	Check: kubectl logs deploy/core-ui -n open5gs